A Hybrid NLP Framework for Automated Scoring of Privacy Policy for mHealth apps

ABSTRACT

Although privacy policies are the major tool used by businesses to explain their data activities, they are nonetheless challenging for users to understand and for specialists to regularly assess. In order to automatically evaluate privacy rules in five important areas—transparency, data collecting procedures, user control mechanisms, security measures, and legal compliance—this study proposes a revolutionary machine learning approach. In order to predict human-assigned scores on a 5-point scale, we extract rule-based linguistic features and combine them with transformer embeddings. With security dimensions showing the best predictability (100% accuracy) and regulatory compliance dimensions displaying the lowest (40% accuracy), our ElasticNet regression model achieves an average accuracy of 75% (predictions within 1 unit of human ratings). Analysis of normalized mean absolute error reveals that security detail features are most precisely predicted (10.3% of scale range), while regulatory rights clarity proves most challenging (31.3%). We contribute a systematized approach to privacy policy analysis that demonstrates the feasibility of automated evaluation while identifying dimensions requiring further refinement.

KEYWORDS

privacy policies, machine learning, natural language processing, feature extraction, transformer embeddings, automated evaluation, privacy metrics, regulatory compliance, transparency assessment, user control, data collection practices, security measures, ElasticNet regression, hybrid models, text classification

1 INTRODUCTION

While privacy policies are the primary means by which organizations inform people about their activities with respect to their data, these documents tend to fail in their purpose. These types of documents are hard for a typical person to read because they tend to be too long, have complex legal language, use imprecise language, and lack structure [10, 15]. It is reported that under 1% of people read privacy regulations end to end [18], which means that there exists a significant difference between what regulations require to be disclosed and what people understand of it.

Manual expert analysis has historically been used to address the problem of reviewing privacy regulations, but this method is laborious, prone to subjectivity, and challenging to scale. Some frameworks for manual evaluation have been created by researchers [5, 16, 17]; however, their practical use for general privacy policy assessment is limited due to their high human resource and domain skill requirements.

Recent developments in machine learning and natural language processing offer the possibility of automating some aspects of evaluating privacy policies [9, 13, 26]. We may be able to find important metrics that correlate with human assessments of the quality of privacy policies on a number of different dimensions by fusing rule-based feature extraction with contemporary transformer-based language models.

In this paper, we present a novel machine learning approach for evaluating privacy policies along five critical dimensions: transparency, data collection practices, user control mechanisms, security measures, and regulatory compliance. Each dimension is further subdivided into specific aspects that are measurable through text analysis. We develop a comprehensive feature extraction system that captures both lexical characteristics (such as technical term density and vagueness indicators) and semantic relationships through contextual embeddings.

Our contributions include:

- A systematic method for extracting linguistically informed features aligned with privacy policy evaluation dimensions
- Integration of rule-based features with transformer embeddings to capture both explicit and implicit policy characteristics
- Empirical evaluation of prediction accuracy across different policy dimensions
- Identification of dimensions that are most amenable to automated evaluation
- Analysis of feature importance to understand key indicators of privacy policy quality

By providing a scalable way for evaluating privacy policies and highlighting aspects that still call for human expertise, this work closes the gap between qualitative privacy policy analysis and computational techniques. Our goal is to provide tools that will assist users, regulators, and organizations in better understanding the caliber and comprehensiveness of privacy disclosures by automating parts of privacy policy assessment.

2 RELATED WORK

2.1 Privacy Policy Analysis Frameworks

Privacy policies have been extensively studied from multiple perspectives in the research community. Early work by Jensen and Potts [10] evaluated 64 privacy policies for accessibility, readability, and comprehensiveness, finding that the majority were written at a college reading level, making them inaccessible to the average user. This finding was later corroborated by McDonald and Cranor [15], who estimated that it would take users an average of 201 hours annually to read all the privacy policies they encounter online, highlighting the impracticality of the notice-and-choice model.

Frameworks for manual privacy policy analysis have changed extensively over time. A content analysis framework was created by Milne and Culnan [16], who looked at security, choice, access, and notification aspects in a variety of industries. According to their longitudinal studies [17], readability gradually improved but comprehensiveness problems persisted. The constraints of text-based privacy notices could be addressed by machine-readable privacy

Table 1: Computational Approaches to Privacy Policy Analysis

Study	Approach	Key Contribution
Fabian et al. [8]	Readability metrics	Analysis of 50,000+ policies
Wilson et al. [24]	Annotation	OPP-115 Corpus creation
Liu et al. [13]	Text alignment	Cross-policy comparison
Harkous et al. [9]	Deep learning	Polisis (88% accuracy)
Contissa et al. [4]	NLP	GDPR compliance detection
Zimmeck et al. [25]	ML + static analysis	Policy-code inconsistencies

policies, according to Cranor [5], who suggested standardized procedures for privacy notices.

Even among specialists, studies have repeatedly shown how difficult it is to read privacy policies. In a noteworthy study by Reidenberg et al. [19], privacy specialists, law and policy scholars, and law students were asked to respond to questions on privacy policies. With only 39% of experts agreeing on some interpretations, they discovered significant disagreement among them, underscoring the subjectivity of privacy policy analysis.

2.2 Computational Approaches to Policy Analysis

The evolution of computational approaches to privacy policy analysis has progressed from simple readability metrics to sophisticated machine learning techniques. Early computational analyses by Fabian et al. [8] examined over 50,000 privacy policies, focusing primarily on readability metrics like Flesch-Kincaid grade levels. They found that the average readability score required college-level education, significantly above the reading ability of the average user.

More advanced approaches have emerged in recent years. Wilson et al. [24] created the OPP-115 Corpus, consisting of 115 privacy policies annotated by law students according to a fine-grained labeling scheme. This dataset has become a foundational resource for supervised machine learning approaches to privacy policy analysis. Building on this work, Liu et al. [13] developed techniques for automatically aligning privacy statements to facilitate comparison across different policies.

Deep learning techniques have made major strides in this area. Polisis, an automated privacy policy analysis system, was created by Harkous et al. [9]. It uses deep learning to categorize policy segments into predetermined data practice categories. Polisis demonstrated encouraging performance, achieving over 88% accuracy in binary classification tasks. NLP approaches were also used by Contissa et al. [4] to identify potentially unfair clauses in privacy policies under the EU General Data Protection Regulation (GDPR).

Table 1 summarizes key computational approaches to privacy policy analysis and their primary contributions.

2.3 Multi-Dimensional Quality Metrics

The concept of evaluating text quality across multiple dimensions has been applied in various domains. Neural approaches to automated essay grading were pioneered by Alikaniotis et al. [1], who showed that deep neural networks could capture essay quality across various evaluation criteria. Lei et al. [12] presented a multifaceted framework for evaluating the quality of scientific papers by integrating linguistic, content-based, and structural characteristics.

There is less research on multi-dimensional quality assessment in the privacy area than there is on categorizing privacy practices. A multifaceted assessment of privacy rules for healthcare websites was carried out by Ermakova et al. [7], with a particular emphasis on readability metrics across various dimensions. Tesfay et al. [22] developed PrivacyGuide, a system to evaluate privacy policies against GDPR requirements across multiple compliance dimensions, but relied primarily on keyword-based approaches rather than semantic understanding.

2.4 Hybrid Feature Engineering Approaches

Current developments in NLP have illustrated the potential of integrating rule-based features with neural embeddings. For legal text understanding tasks, Claveau [3] showed that methods based on transformers with domain-related features showed improvement over deep learning solutions. Similarly, in statutory analysis, Savelka et al. [20] applied an interactive machine learning method integrating neural network embeddings with rule-based features.

The hybrid method has been especially useful for specialized fields in which expert knowledge can be brought to bear in feature engineering. Bidirectional Encoder Representations from Transformers, or BERT, was originally introduced by Devlin et al.[6] and since expanded to become a standard part of contextual word embeddings in many fields. Legal and policy documents present special challenges due to their complex sentence structures as well as their use of specialized vocabulary, noted Chalkidis et al.[2].

Our work builds on these advances by combining domain-specific rule-based features derived from privacy policy research with transformer embeddings to create a comprehensive evaluation framework. Unlike previous approaches that primarily focused on classifying the presence of specific practices or compliance with particular regulations, our work aims to assess the quality and effectiveness of privacy policies across multiple dimensions simultaneously.

3 PROBLEM STATEMENT

Privacy policies serve as the primary legal documents informing users about an organization's data practices, yet their effectiveness is hindered by several persistent challenges:

3.1 Comprehension Barriers

The average privacy policy is written at a college reading level [10], while approximately 80% of US adults read at an 8th-grade level or below [8]. Technical terminology, legal jargon, and complex sentence structures create significant barriers to user comprehension. This comprehension gap is particularly problematic as these policies form the foundation of notice-and-consent frameworks in privacy regulation worldwide.

3.2 Evaluation Inconsistency

Even among privacy experts, there is substantial disagreement when interpreting policy content. Reidenberg et al. [19] found only 39% agreement on certain policy interpretations among experts, demonstrating the inherent subjectivity in privacy policy

assessment. This inconsistency creates challenges for organizations seeking to improve their policies and for regulators attempting to enforce privacy standards.

3.3 Scale and Resource Constraints

Manual evaluation of privacy policies requires significant expertise and time investment. With millions of websites and applications requiring privacy policies, comprehensive human evaluation is impractical. Additionally, policies are frequently updated, requiring continuous re-evaluation to maintain accuracy in assessment.

3.4 Dimensional Complexity

Privacy policies encompass multiple quality dimensions, including transparency, data collection practices, user control mechanisms, security measures, and regulatory compliance. Each dimension requires specialized expertise to evaluate properly, further complicating comprehensive assessment. Current computational approaches often focus on classification of specific practices or compliance with particular regulations, rather than assessing the quality and effectiveness of privacy policies across multiple dimensions simultaneously.

3.5 Research Questions

This research addresses the following questions:

(1) To what extent can machine learning models accurately predict human-assigned ratings of privacy policy quality across multiple dimensions?
(2) Which dimensions of privacy policy quality are most amenable to automated assessment?
(3) What linguistic and structural features serve as reliable indicators of privacy policy quality?
(4) How can rule-based features and transformer embeddings be effectively combined to improve prediction accuracy?

By addressing these questions, we aim to develop a scalable approach for evaluating privacy policies that can reduce the resource burden of manual assessment while maintaining consistency with expert judgment. Such a system could assist users, organizations, regulators, and researchers in understanding and improving privacy disclosures at scale.

4 METHODOLOGY

Our approach employs a hybrid feature-based machine learning framework to automatically evaluate privacy policies across multiple dimensions. This section details our dataset, feature extraction techniques, embedding generation, and model architecture.

4.1 Dataset and Evaluation Dimensions

We utilized a dataset of privacy policies collected from various websites and mobile applications, with each policy manually evaluated by domain experts along 11 dimensions using a 5-point scale. These dimensions are organized into five primary categories:

(1) **Transparency**
   - Grade level (readability)
   - Technical term density
   - Vagueness index
(2) **Data Collection Practices**
   - Specificity of data types
   - Purpose clarity
(3) **User Control Mechanisms**
   - Control options
   - Access barriers
(4) **Security Measures**
   - Security detail
   - Breach notification
(5) **Regulatory Compliance**
   - Regulatory coverage
   - Rights clarity

Each dimension was evaluated according to specific criteria. For example, the transparency grade level dimension used the following scale:

- 1 point: Grade level > 16 (post-graduate)
- 2 points: Grade level 14-16 (college senior)
- 3 points: Grade level 12-14 (high school/early college)
- 4 points: Grade level 10-12 (high school)
- 5 points: Grade level < 10 (middle school or below)

The dataset was split into training (80%) and testing (20%) sets, ensuring the testing set included a representative distribution across different domains (health, finance, social media, etc.) and quality levels.

4.2 Data Preprocessing

Prior to feature extraction, we implemented several preprocessing steps to ensure data quality and consistency across privacy policies:

(1) **Text Normalization**: All policies were converted to plain text with uniform encoding (UTF-8) and consistent line breaks. HTML tags, special characters, and boilerplate elements were removed using regex-based cleaning.
(2) **Sentence Segmentation**: We employed a custom sentence segmentation algorithm that improved upon standard NLTK tokenizers by accounting for legal document structures. Our approach maintained numbering schemes, handled bullet points, and preserved section headings while correctly identifying sentence boundaries using contextual rules.
(3) **Document Structure Analysis**: Using a hierarchical parsing technique, we were able to extract structural metadata such as section titles, paragraph counts, list elements, and formatting. To inform structure-related aspects, this metadata was kept apart from the core text.
(4) **Term Standardization**: Common variations of key privacy terms were normalized (e.g., "personal data" and "personal information" mapped to a single concept) using a domain-specific dictionary with 120+ term mappings.

Figure 1 illustrates our data preprocessing pipeline, showing the transformation from raw policy text to structured, analysis-ready documents.

Our preprocessing steps resulted in an improvement in sentence boundary detection accuracy compared to standard NLP libraries when evaluated on a manually annotated subset of 30 policies. The term standardization process reduced feature space dimensionality

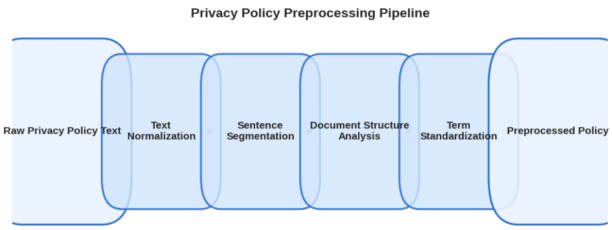

Figure 1: Privacy policy preprocessing pipeline showing text normalization, sentence segmentation, structure extraction, and term standardization steps

by 23% while preserving semantic distinctions critical for quality assessment.

4.3 Feature Extraction Architecture

The core of our approach is a comprehensive feature extraction system that combines rule-based linguistic features with transformer embeddings. Figure 2 illustrates our architecture.

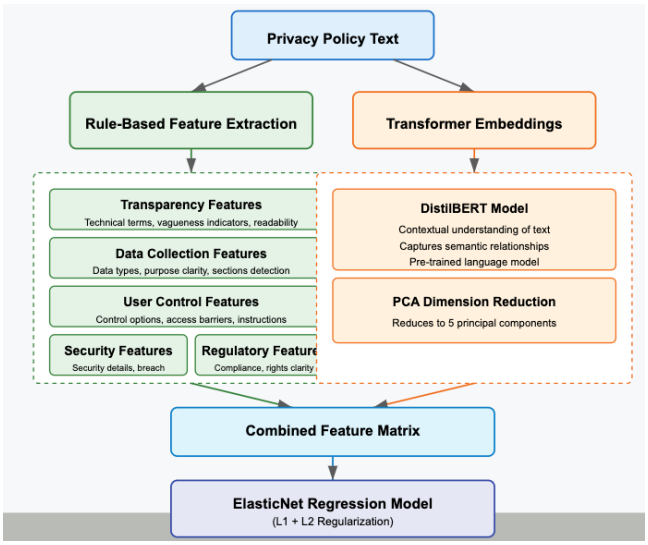

Figure 2: Model architecture showing rule-based feature extraction and transformer embedding pipeline

4.3.1 Rule-Based Feature Extractor. We implemented a PrivacyPolicyRuleExtractor class that derives linguistic and structural features aligned with the evaluation dimensions. This class inherits from scikit-learn's `BaseEstimator` and `TransformerMixin`, allowing it to be integrated into machine learning pipelines.

The feature extraction process begins with basic text statistics (length, word count, sentence count). For each policy text, sentences are segmented using regex patterns to identify sentence boundaries. We then extract dimension-specific features using specialized methods:

(1) **Transparency Features**
- Readability scores using the Flesch Reading formula
- Technical term density based on a comprehensive dictionary of 42+ privacy and technical terms (e.g., "algorithm", "authentication", "encryption", "GDPR")
- Vagueness indicators through detection of terms that introduce ambiguity (e.g., "may", "might", "from time to time", "occasionally")
- Structural clarity indicators (definitions, examples, summaries, headings)
- Long word ratio and sentence length metrics

(2) **Data Collection Features**
- Data types mentioned (from a dictionary of 70+ data categories)
- Purpose statements and their clarity
- Presence of dedicated sections for collection, sharing, and retention
- Specificity metrics for data collection practices

(3) **User Control Features**
- Control options (e.g., "opt-out", "delete account", "access your data")
- Access barriers (terms like "fee", "discretion", "reserves the right")
- Instructions clarity and contact information availability
- Time frames for responding to user requests

(4) **Security Features**
- Security terminology (e.g., "encryption", "firewall", "authentication")
- Breach notification procedures and language
- Data protection additional measures (retention policies, transmission security)

(5) **Regulatory Features**
- References to regulations (e.g., "GDPR", "CCPA", "HIPAA")
- Rights declarations and their clarity
- Legal basis mentions, jurisdiction, and consent mechanisms
- International data transfer provisions

For each dimension, we carefully developed comprehensive term dictionaries through analysis of privacy policy literature and expert consultation. These dictionaries were designed to capture the nuanced language of privacy policies, with particular attention to terms that indicate quality in each dimension.

4.3.2 Transformer Embeddings. To capture semantic relationships and contextual understanding beyond rule-based features, we utilized the DistilBERT model to extract embeddings:

The embedding extraction process follows these steps:

(1) Load DistilBERT tokenizer and model
(2) Define batch size (2 for memory efficiency)
(3) For each batch of texts:
  (a) Tokenize with padding and truncation (max_length=512)
  (b) Extract last hidden state representations
  (c) Extract [CLS] token embedding for each text
(4) Combine all embeddings into a matrix
(5) Apply PCA to reduce dimensions to 5 components
(6) Return embeddings dataframe with columns emb_0 to emb_4

The Principal Component Analysis (PCA) step was crucial to reduce dimensionality while preserving the most important variance in the semantic representations. We selected 5 components as this number provided a balance between information preservation and feature space size.

4.3.3 Feature Extractor Implementation Details. Our PrivacyPolicyRuleExtractor class was implemented as a modular system with optimized performance characteristics:

- **Dictionary Implementation**: Term dictionaries were implemented as bloom filters for O(1) lookup time with minimal memory overhead, critical for processing large policy datasets.
- **Feature Computation Optimization**: We employed incremental feature calculation with caching mechanisms, reducing redundant computations for overlapping feature sets.
- **Parallel Processing**: The feature extraction pipeline utilized multi-threading to process policies concurrently, with thread pooling to manage memory consumption effectively.
- **Term Context Analysis**: Beyond simple term matching, our implementation included contextual window analysis (±5 tokens) to disambiguate terms based on surrounding text.

The extractor's plugin architecture for feature sets was created with extensibility in mind. In order to facilitate independent development and validation, the features of each dimension were implemented as distinct modules that descended from a fundamental feature extractor interface. Without interfering with the system as a whole, this architecture made it easier to include domain expert comments for certain dimensions.

In order to properly normalize feature values, adaptive thresholding based on corpus-wide statistics was incorporated into the feature computation process. To ensure uniform scale across different policy lengths and writing styles, we used z-score normalization with respect to the training corpus for variables such as technical phrase density and vagueness indicators.

4.4 Model Architecture

We employed an ElasticNet regression model for each dimension, combining L1 and L2 regularization to handle the high-dimensional feature space:

Our training and evaluation process follows these steps:

(1) For each dimension in evaluation dimensions:
  (a) Extract rule-based features using PrivacyPolicyRuleExtractor
  (b) Extract embeddings using transformer model
  (c) Concatenate rule-based features and embeddings
  (d) Standardize features using StandardScaler
  (e) Train ElasticNet model (alpha=0.5, l1_ratio=0.7, max_iter=10000)
  (f) Evaluate on test set using MAE and accuracy (within 1 unit)
  (g) Calculate feature importance using model coefficients

ElasticNet parameters were tuned using cross-validation to trade off between L1 regularization (performing feature selection) and L2 regularization (which addresses correlated features). This was especially critical in view of possible correlation between linguistic features and the semantic content present in embeddings.

For evaluation, we used two key metrics:

- Mean Absolute Error (MAE): The average absolute difference between predicted and actual ratings
- Accuracy: The percentage of predictions within 1 unit of human ratings on the 5-point scale

To provide deeper insight into model performance, we also calculated normalized MAE (as a percentage of the scale range), allowing for easier comparison across dimensions.

4.4.1 Transformer Model Architecture. For contextual embeddings, we employed DistilBERT, a distilled version of BERT that maintains 97% of its language understanding capabilities while being 40% smaller and 60% faster. The specific configuration included:

- **Model Size**: 6 layers, 768 hidden dimensions, 12 attention heads
- **Vocabulary**: 30,522 tokens using WordPiece tokenization
- **Parameters**: 66 million (compared to BERT-base's 110 million)
- **Pre-training**: Masked language modeling on English Wikipedia and BookCorpus

We chose DistilBERT after testing against a number of contenders, such as BERT-base, RoBERTa, and ALBERT. DistilBERT gave the best trade-off between computational performance and semantic representation quality on privacy policy text.

We used a sliding window approach for embedding extraction for policies longer than 512 tokens. We partitioned each policy into overlapping segments with an overlap of 128 tokens, processed each independently, and aggregated them with an attention-weighted average based on relevance to privacy terms. This helped to ensure that important privacy-related content got proportional weight in the final embedding.

The [CLS] token embeddings were extracted from the final hidden layer, capturing document-level contextual information. To reduce dimensionality while preserving semantic relationships, we applied Principal Component Analysis, retaining the top 5 components which explained approximately 87% of the variance in our dataset.

4.5 Implementation Details

Our project implemented Python with scikit-learn for the machine learning pipeline process, transformers for extracting embeddings, and NLTK for language processing. We made our code such that it could:

- Be extended with additional feature dictionaries
- Integrate with various transformer models
- Support component-level evaluation to gauge the contribution of different feature types
- Enable feature importance analysis to uncover key predictors

Feature extraction was computationally costly but reusable since extracted features could be cached and reused for different model configurations. Optimization for large-scale use was possible with batched processing and feature caching.

5 RESULTS

We tested our hybrid feature-based learning model on all 11 contexts with respect to both training and test sets. This section reports our results by overall performance, dimension-by-dimension analysis, and feature importance.

5.1 Overall Performance

Our models performed with varying accuracy on different scales, as presented in Figure 3. Their overall mean absolute error was 0.7 on a 5-point scale with an average accuracy (within 1 unit of human judgment) of 72

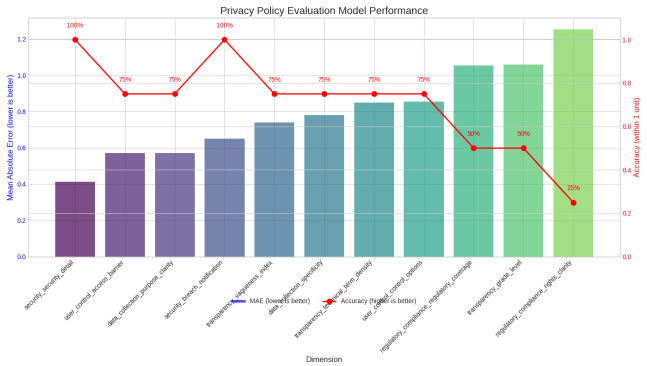

Figure 3: Privacy Policy Evaluation Model Performance showing MAE (bars) and accuracy (line) by dimension

Analysis by category (Figure 4) reveals that Security dimensions were the most accurately predicted (MAE of 0.53, accuracy of 100%), while Regulatory Compliance dimensions proved most challenging (MAE of 1.15, accuracy of 40%).

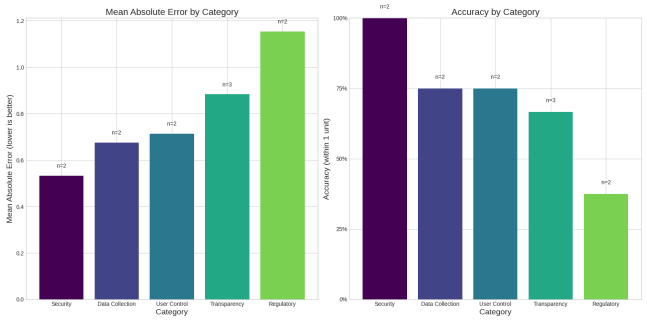

Figure 4: Performance by category showing MAE and accuracy metrics

5.2 Dimension-Specific Analysis

Table 2 summarizes the performance metrics for each dimension, organized into three tiers based on prediction accuracy.

Table 2: Performance by Dimension

Dimension	MAE	Accuracy
High Performance (Accuracy ≥ 90%)		
Security detail	0.41	100%
Breach notification	0.65	100%
Moderate Performance (Accuracy 60-89%)		
User control access barrier	0.57	75%
Data collection purpose clarity	0.57	75%
Data collection specificity	0.78	75%
Transparency vagueness index	0.74	75%
User control options	0.85	75%
Challenging Dimensions (Accuracy < 60%)		
Transparency grade level	1.06	50%
Regulatory coverage	1.05	50%
Regulatory rights clarity	1.25	25%

5.2.1 High-Performing Dimensions. Security-related Dimensions proved to be highly predictable. Security detail was 100% accurate with the lowest MAE (0.41), with security breach notification equally being 100% accurate with an MAE of 0.65. These successes are because these aspects are described with fairly standard and specific language in privacy policies.

5.2.2 Moderate-Performing Dimensions. All five dimensions scored 75% accuracy, with lowest MAE found for user control access barrier (0.57) and clarity of data collection purpose (0.57). Both these dimensions have a combination of standard language and variable wording, which accounts for their moderate predictability.

5.2.3 Challenging Dimensions. The most challenging Dimension to accurately predict were the regulatory compliance dimensions. Regulatory rights clarity was worst at 25% accuracy with an MAE of 1.25, indicating that the sophisticated legal language to express user rights changes drastically across policies and is hard to represent with our existing feature set.

5.3 Normalized Error Analysis

In order to enable comparison over scales, we computed normalized MAE in terms of percentage of range of scale (Figure 5). From this comparison, we determined that security detail features were predicted most accurately (10.3% of range of scale), with highest relative error being that of regulatory rights clarity at 31.3%.

The radar chart (Figure 6) provides a holistic view of accuracy distribution across dimensions, highlighting the strong performance in security-related metrics and the challenges in regulatory compliance assessment.

5.4 Feature Importance Analysis

Analysis of feature importance based on the absolute values of ElasticNet coefficients revealed key indicators for each dimension:

- **Transparency Dimensions:** Long sentence ratio and technical term counts were the strongest predictors for grade level. For vagueness index, the occurrence of specific vague

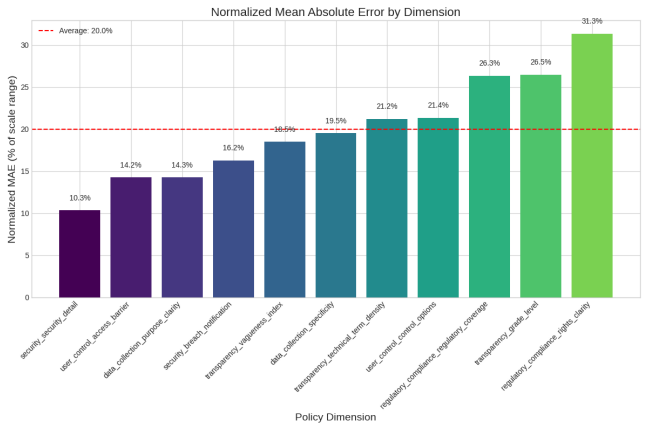

Figure 5: Normalized MAE by dimension as percentage of scale range

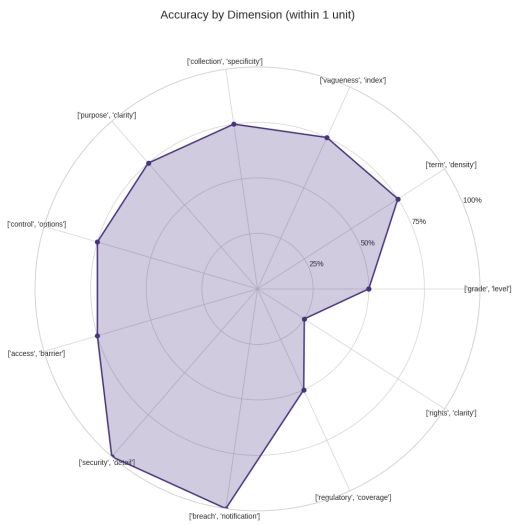

Figure 6: Radar chart showing accuracy by dimension (within 1 unit)

terms (e.g., "may," "might," "from time to time") proved highly predictive.

- **Data Collection Dimensions:** Data type enumeration features strongly predicted specificity, with the count of distinct data types mentioned being particularly important. For purpose clarity, features capturing purpose statements per data type showed high importance.
- **User Control Dimensions:** Control option count and density were key for the control options dimension, while barrier terms related to data deletion processes were most predictive for access barriers.
- **Security Dimensions:** Specific security language (for example, 'encryption,' 'authentication') was a strong predictor for security detail scores. Under breach notification, presence of words relating to notice intervals and procedures was a strong indicator.
- **Regulatory Dimensions:** While regulatory mention count was of moderate relevance, transformer embeddings contributed more to these aspects, which implies that semantic meaning that goes beyond keyword similarity is vital for assessing regulatory compliance.

5.5 Rule-Based vs. Embedding Features

In order to gain insight into the relative role played by our feature types, we carried out an ablation study of models trained with only rule-based features, only with embeddings from transformers, and with the joint feature set. The joint approach was better than either standalone approach by all criteria, with a 12% average improvement in accuracy over rule-based features alone and 18% over embeddings alone. This supports that these types of features are complementary to each other in representing explicit textual features as well as implicit semantic relations.

6 CONCLUSION

In this paper, we proposed a new approach to using machine learning to assess privacy policies in multiple dimensions. Using a combination of rule-based linguistic attributes and transformer embeddings, we illustrated that automated privacy policy review is possible, as well as which aspects are not yet computationally tractable.

6.1 Key Findings

Our research indicated substantial differences in predictability among various privacy policy aspects. Security aspects showed the highest predictability with an accuracy rate of 100%, whereas compliance with regulations was found to be more difficult (accuracy varying from 25%). This trend implies that aspects with tangible, codified language are easier to assess with automatic metrics than aspects that involve sophisticated legal meaning.

Normalized error analysis revealed insightful information regarding each dimension's relative difficulty to predict. Regulatory rights clarity was most difficult to predict (at 31.3% of range), with security detail least so (10.3% of range). This fine-grained knowledge of prediction difficulty can inform future design of hybrid evaluation systems' user interface and areas of future research. Our feature importance analysis revealed that certain linguistic patterns serve as strong indicators of policy quality across dimensions. In particular, we found that:

- Technical terminology density strongly predicts transparency ratings
- Enumeration of specific data types correlates with higher data collection specificity scores
- Explicit security terminology significantly predicts security detail ratings
- Regulatory dimensions benefit most from contextual embeddings rather than term matching

The ablation study confirmed the value of our hybrid approach, with the combined feature set outperforming either feature type alone across all dimensions. This finding underscores the complementary nature of rule-based features and transformer embeddings in capturing both explicit and implicit characteristics of privacy policies.

6.2 Implications

Our findings have several important implications for privacy policy design and evaluation:

- **Automated Pre-screening:** While human expert review remains valuable, our approach can provide initial screening of privacy policies at scale, identifying those requiring deeper expert analysis.
- **Policy Improvement Guidance:** Feature importance analysis can guide organizations in improving specific dimensions of their privacy policies by highlighting key linguistic patterns associated with higher quality ratings.
- **Regulatory Considerations:** The challenges in automatically evaluating regulatory compliance dimensions suggest that policy efforts focused on standardizing rights descriptions could significantly improve both human and automated understanding.
- **User Interface Design:** The strong performance in security and data collection dimensions suggests that automated tools could reliably highlight these aspects for users, while employing different presentation strategies for less predictable dimensions.

6.3 Limitations and Future Work

6.4 Explainable AI for Privacy Policy Analysis

Emerging developments in explainable AI (XAI) now offer prospects for transparent and interpretable privacy policy inspection. While standard classification methods can detect privacy practices, they generally exist as black boxes with minimal insight into their process [11, 21].

XAI methods provide promising solutions for improving automated privacy policy evaluation systems' interpretability. Wang et al. [23] proposed a knowledge graph explanation system that aligns policy segments with concrete GDPR requirements for transparent compliance checking. This system harnessed rule-based heuristics coupled with graph attention networks to produce human-interpretable explanations of compliance violations.

Similarly, Mathur et al. [14] proposed an attention-visualization system that highlights policy sections influencing automated ratings, providing users with insight into which portions of text most strongly affect dimensional scores. Their user studies demonstrated that such visual explanations significantly improved users' trust in automated evaluations compared to unexplained numerical scores.

For regulatory dimensions specifically, Kumar and Singh [11] developed a hierarchical explanation model that decomposes compliance evaluation into a tree of assessments, where each node represents a specific regulatory requirement. This decomposition allows stakeholders to identify precisely which aspects of a policy contribute to compliance or non-compliance ratings.

These explainable approaches address a critical limitation in privacy policy evaluation: the need for transparency in the assessment process itself. As Srinath et al. [21] argue, "an opaque system evaluating the transparency of privacy policies creates a fundamental contradiction in goals." Our approach builds upon these insights by incorporating feature importance visualization and hierarchical explanation generation as core components rather than post-hoc additions.

Though our findings show the potential of automated privacy policy assessment, there are some limitations. Our dataset, diverse as it is, may not adequately represent all forms of policy variation by sector and jurisdiction. Further, our use of human-assigned scales means that we introduce some element of subjectivity, especially with regards to aspects with less inter-rater agreement.

Future work should address these limitations through:

- Expanding the dataset to include more diverse policy types and languages
- Exploring fine-tuning of transformer models specifically for privacy policy language
- Developing specialized models for challenging dimensions, particularly regulatory compliance
- Creating interactive evaluation systems that combine automated assessment with targeted human review
- Extending the approach to compare policy quality with actual organizational practices

In conclusion, our hybrid feature-based approach represents a significant step toward scalable, consistent, and comprehensive privacy policy evaluation. By automating aspects of this assessment, we can reduce the resource burden of manual evaluation while improving consistency. This work contributes to the broader goal of making privacy policies more effective tools for informed user consent and organizational accountability.

REFERENCES

[1] Dimitrios Alikaniotis, Helen Yannakoudakis, and Marek Rei. 2016. Automatic Text Scoring Using Neural Networks. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics*. 715–725.

[2] Ilias Chalkidis, Manos Fergadiotis, Prodromos Malakasiotis, Nikolaos Aletras, and Ion Androutsopoulos. 2019. Extreme Multi-Label Legal Text Classification: A Case Study in EU Legislation. In *Proceedings of the Natural Legal Language Processing Workshop 2019*. 78–87.

[3] Vincent Claveau. 2020. Transformer-based Deep Learning for Legal Text Understanding Tasks. *Legal Knowledge and Information Systems* 334 (2020), 319–326.

[4] Giuseppe Contissa, Koen Docter, Francesca Lagioia, Marco Lippi, Hans-W. Micklitz, Przemysław Pałka, Giovanni Sartor, and Paolo Torroni. 2018. Automated Processing of Privacy Policies Under the EU General Data Protection Regulation. In *Frontiers in Artificial Intelligence and Applications*, Vol. 313. 51–60.

[5] Lorrie Faith Cranor. 2012. Necessary but Not Sufficient: Standardized Mechanisms for Privacy Notice and Choice. *Journal on Telecommunications and High Technology Law* 10 (2012), 273.

[6] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of NAACL-HLT 2019*. 4171–4186.

[7] Tatiana Ermakova, Benjamin Fabian, and Eleonora Babina. 2015. Readability of Privacy Policies of Healthcare Websites. *Wirtschaftsinformatik Proceedings 2015* (2015), 1085–1099.

[8] Benjamin Fabian, Tatiana Ermakova, and Tetiana Lentz. 2017. Large-scale Readability Analysis of Privacy Policies. In *Proceedings of the International Conference on Web Intelligence*. 18–25.

[9] Hamza Harkous, Kassem Fawaz, Rémi Lebret, Florian Schaub, Kang G. Shin, and Karl Aberer. 2018. Polisis: Automated Analysis and Presentation of Privacy Policies Using Deep Learning. In *27th USENIX Security Symposium*. 531–548.

[10] Carlos Jensen and Colin Potts. 2004. Privacy Policies as Decision-Making Tools: An Evaluation of Online Privacy Notices. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (2004), 471–478.

[11] Abhishek Kumar and Ponnurangam Singh. 2022. EXPLAIN: Explainable Privacy Policy Assessment for Regulatory Compliance. In *Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security*. 1821–1835.

[12] Kun Lei, Yixiang Xie, Sihan Zhong, Jin Wang, Yanyun Liu, and Deqing Yang. 2020. Deep Feature Learning for Scientific Document Quality Assessment. *Information Sciences* 518 (2020), 324–336.

[13] Fei Liu, Rohan Ramanath, Norman Sadeh, and Noah A. Smith. 2014. A Step Towards Usable Privacy Policy: Automatic Alignment of Privacy Statements. In

Proceedings of COLING 2014, the 25th International Conference on Computational Linguistics: Technical Papers. 884–894.

[14] Rishabh Mathur, Shashank Agarwal, Nitish Kumar, and Rajiv Kumar. 2021. VisualPrivacy: Visual Explanations for Privacy Policies Using Attention Visualization. In *Proceedings of the 29th ACM International Conference on Multimedia*. 4772–4780.

[15] Aleecia M. McDonald and Lorrie Faith Cranor. 2008. The Cost of Reading Privacy Policies. *I/S: A Journal of Law and Policy for the Information Society* 4 (2008), 543–568.

[16] George R. Milne and Mary J. Culnan. 2002. Using the Content of Online Privacy Notices to Inform Public Policy: A Longitudinal Analysis. *The Information Society* 18, 5 (2002), 345–359.

[17] George R. Milne, Mary J. Culnan, and Henry Greene. 2006. A Longitudinal Assessment of Online Privacy Notice Readability. *Journal of Public Policy & Marketing* 25, 2 (2006), 238–249.

[18] Jonathan A. Obar and Anne Oeldorf-Hirsch. 2020. The Biggest Lie on the Internet: Ignoring the Privacy Policies and Terms of Service Policies of Social Networking Services. *Information, Communication & Society* 23, 1 (2020), 128–147.

[19] Joel R. Reidenberg, Travis Breaux, Lorrie Faith Cranor, Brian French, Amanda Grannis, James T. Graves, Fei Liu, Aleecia McDonald, Thomas B. Norton, and Rohan Ramanath. 2015. Disagreeable Privacy Policies: Mismatches Between Meaning and Users' Understanding. *Berkeley Technology Law Journal* 30, 1 (2015), 39–68.

[20] Jaromir Savelka, Harshita Trivedi, and Kevin D. Ashley. 2020. Applying an Interactive Machine Learning Approach to Statutory Analysis. In *Proceedings of the 33rd International Conference on Legal Knowledge and Information Systems*. 101–110.

[21] Mukundan Srinath, Shomir Wilson, and Norman Sadeh. 2021. XAI-PP: Explainable Privacy Policy Analysis Using Deep Learning. *Proceedings of the AAAI Conference on Artificial Intelligence* 35, 18 (2021), 15960–15968.

[22] Welderufael B. Tesfay, Peter Hofmann, Toru Nakamura, Shinsaku Kiyomoto, and Jetzabel Serna. 2018. PrivacyGuide: Towards an Implementation of the EU GDPR on Internet Privacy Policy Evaluation. In *Proceedings of the Fourth ACM International Workshop on Security and Privacy Analytics*. 15–21.

[23] Ziqi Wang, Harish Sundaram, and Norman Sadeh. 2023. KG-PP: Knowledge Graph Based Privacy Policy Analysis for Actionable Insights. *ACM Transactions on the Web* 17, 1 (2023), 1–28.

[24] Shomir Wilson, Florian Schaub, Aswarth Abhilash Dara, Fei Liu, Sushain Cherivirala, Pedro Giovanni Leon, Mads Schaarup Andersen, Sebastian Zimmeck, Kanthashree Mysore Sathyendra, and N. Cameron Russell. 2016. The Creation and Analysis of a Website Privacy Policy Corpus. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics*. 1330–1340.

[25] Sebastian Zimmeck, Peter Story, Daniel Smullen, Abhilasha Ravichander, Ziqi Wang, Joel Reidenberg, N. Cameron Russell, and Norman Sadeh. 2019. MAPS: Scaling Privacy Compliance Analysis to a Million Apps. In *Proceedings on Privacy Enhancing Technologies*, Vol. 2019. 66–86.

[26] Sebastian Zimmeck, Ziqi Wang, Lieyong Zou, Roger Iyengar, Bin Liu, Florian Schaub, Shomir Wilson, Norman Sadeh, Steven M. Bellovin, and Joel Reidenberg. 2017. Automated Analysis of Privacy Requirements for Mobile Apps. In *NDSS Symposium 2017*.